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Preliminary Appraisal

Limmattal Bike Highway

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1 Introduction

1.1 Base situation

1.1.1 The Limmattal

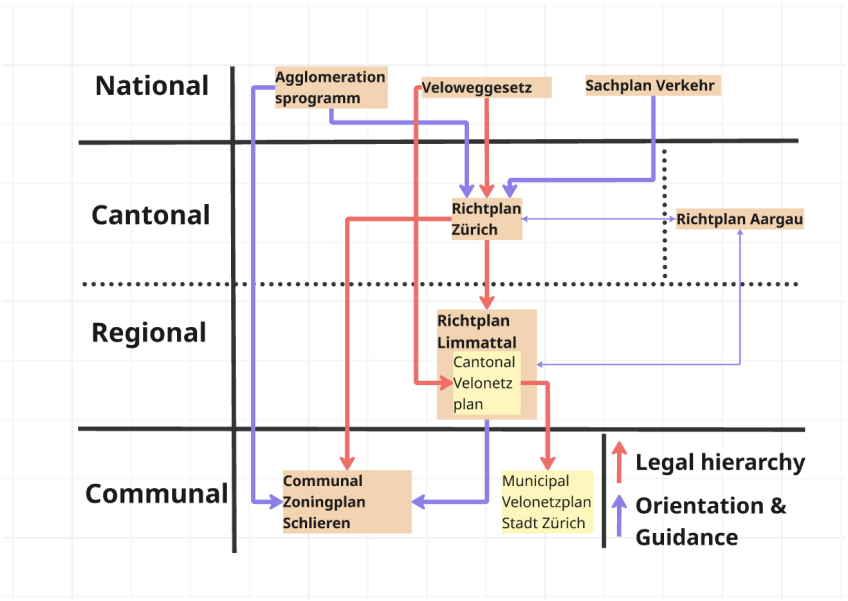
The Limmattal is a densely populated area located between the city of Zurich and Baden, home to approximately 85'000 inhabitants. Extending over 12 km from the cantonal border to the outskirts of Zurich, it serves as an important commuting zone with around 100'000 daily trips between the Limmattal and the city alone (Kanton Zürich, 2017). The combination of high travel demand towards the urban centre, relatively short distances, and the flat topography along the Limmat river make the area particularly suitable for bicycle commuting. However, despite this favourable location, an attractive bicycle infrastructure is lacking. As a result, around 64% of the daily distance traveled within the region is still covered by car (Kanton Zürich, 2017).

1.1.2 Legal framework

Looking at the legal framework surrounding the bicycle highway, at the federal level, the influential law the Veloweggesetz stimulates a change in such underdeveloped bike infrastructure.(Vel, 2022). The Law of 2021 mandates the cantons to plan for a continuous and uninterrupted network of cycle paths to connect the population. These bicycle paths need to be designed for safety, directness, and separation from other traffic and pedestrians. Importantly, the cantons need to enact plans until 2027 and must implement the planned projects until 2042. Therefore, there is an imperative to move on quickly with such bicycle projects. On a cantonal level, the Veloweggesetz mandates a Velonetzplan, *bicycle network plan*, which would plan the cantonal network. In this case the canton of Zurich integrated the Velonetzplan in the regional Richtplan, *structure plan*, of the Limmat-Valley. The unique extent of the bicycle highway from the city border of Zurich towards the cantonal border signifies that two other legal processes are at work. On the one hand, the city of Zurich plans its own network in a communal Velonetzplan; meanwhile, the cantonal Velonetzplan, the regional Richtplan, and the cantonal Richtplan all need to orient themselves to the Aargau cantonal Richtplan (Kanton Aargau, 2025). Therefore, the routing of the bicycle highway in the limmattal needs to consider the planned bicycle routes of the city of Zurich as well as the canton of Aargau to plan for an uninterrupted and continuous network as mandated by the Veloweggesetz. Returning to the national level, the Agglomerationsprogramm, which is currently in its 5. Generation can provide

funding for the bicycle highway to the canton (Kanton Zürich, 2025). Meanwhile, also on the national level, the Sachplan Verkehr provides guidance for the cantonal Richtplan to plan for Velobahnen which connect inner cities with the agglomeration, like our project (ARE, 2021). On the communal level, the Zoningplan of the involved municipalities: Schlieren, Dietikon and Unterengstringen must allow yielding the land for the Velobahn. Especially the Zoningplan of Schlieren must be revised to allow this, which can happen in 2026 the earliest (Kanton Zürich, 2025).

Figure 1: Legal framework of the instruments invloved

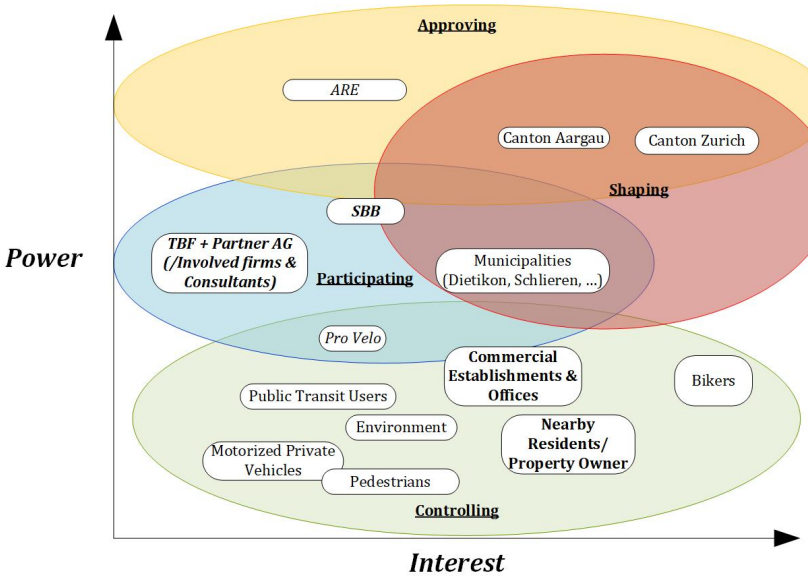


1.1.3 Stakeholders

The proposed bike highway in the Limattal stretches over a large areas, encompassing two cantons (Aargau and Zurich) and multiple municipalities. This project might be preceded and developed by the larger Veloweggesetz and Velonetzplan, but on a more granular level it needs to fit an amalgamation of needs and concerns of different actors. A stakeholder analysis aims to incorporate involved stakeholders in an effective and just manner, based on the level of involvement in the project. Such an analysis is crucial for translating strategic plans to actionable proposals (top-down) or local needs to the larger urban fabric (bottom-up). In addition, it aims at spotting risks and key players ex ante, to make sure the course of the infrastructure planning unfolds in a coordinated and satisfying manner. A stakeholder analysis has been done in this report to understand the complexity and nature of this infrastructure proposal, as can be read in Appendix B. In figure 2 the results are displayed in a Power-Interest matrix. For the model it is

important to note that the economic viability is of high strategic importance for shaping stakeholders. On the other hand, the behavior of controlling stakeholders determines the aggregated societal costs.

Figure 2: Stakeholder Analysis key figure



1.2 Expected change

As streets and public transport systems become increasingly congested, a growing number of people are turning to the bicycle as an alternative mode of travel. In Switzerland, for example, the total distance traveled by bicycle rose by around 13% between 2010 and 2015, indicating a clear upward trend in cycling participation. Although cycling is theoretically possible almost everywhere, in practice cyclists often have to share streets with cars, trucks, and busses, which substantially reduces perceived safety. This situation may be acceptable for short local trips, but it becomes increasingly problematic on medium-distance journeys of approximately 5 to 15 kilometres, the typical commuting distance from suburban agglomerations to city centres. Consequently, many commuters continue to rely on cars or public transport, leaving considerable untapped potential for cycling. To fully utilize this potential, infrastructural measures are required, as cyclists tend to prioritize high safety standards and minimal physical effort. An attractive cycling network therefore depends on routes that are physically separated from motorized traffic, highly direct, and characterized by a minimal number of stops, which are central principles in the development of bicycle highways.

1.2.1 Societal Trends

Studying the bottom-up processes concerning the bike highway, important societal trends have been identified. The sustainable mobility change has led more people to think about how their own mobility impacts climate change and their communities. This dynamic could be exemplified by the residents of the nearby city of Zurich, who voted to fund a bicycle highway within the city and subsequently voted again for increased funding. The critical mass movement has further shown for many years that expanding the bike network has garnered outspoken support from the population.

1.3 Infrastructure expectations and proposal

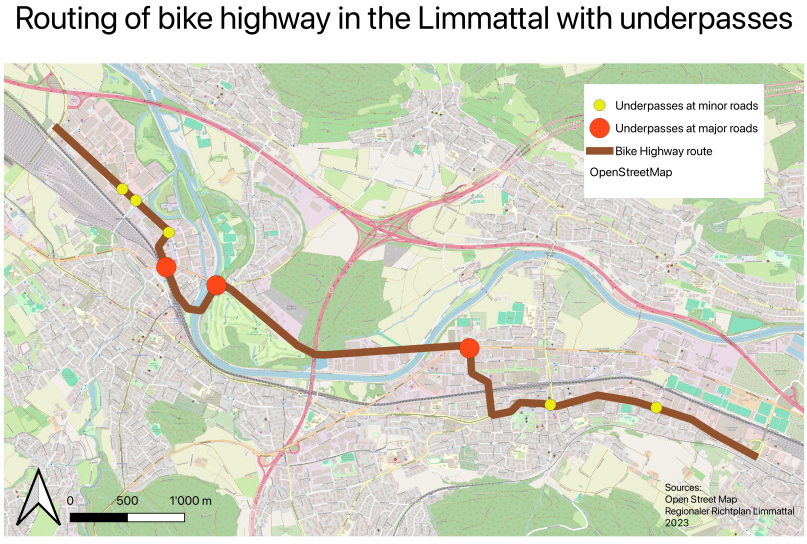
These initiatives have set the catalyst for the development of a bike highway in the Limmattal Valley. In this paragraph, the expectations of such a project and its proposal are discussed.

Expectations: The overarching expectation is to create an infrastructure that meaningfully encourages cycling as a sustainable, efficient, and attractive mode of transport, particularly on medium-distance commuter trips of roughly 8–15 kilometres. The aim is to provide a mobility option that is safe, direct, and not unnecessarily exhausting, enabling users to travel at consistently high and predictable speeds while feeling well-protected from motorised traffic. By offering a compelling alternative to the private car for these medium-range journeys, the corridor is expected to increase the bicycle modal share and simultaneously decrease car usage along the route. Additionally, the project seeks to reduce emissions, improve public health through active mobility, and lower accident risks and associated costs compared to current conditions. Outcomes considered unacceptable include any increase in emissions or accident rates relative to today, or the creation of infrastructure that fails to deliver a safety or travel-time advantage over the existing network. In the long term, success will be reflected in a corridor that supports growing cycling demand, ensures a comfortably high level of safety and continuity, and positions cycling as a competitive alternative to both private cars and public transport within the Limmattal region.

Proposed Infrastructure: The proposed bike highway is an approximately 8.5 km long, route connecting Altstetten and the cantonal border at Dietikon. It is designed to offer a continuous, comfortable and fast travel experience for cyclists, following Vel (2022). The corridor follows the Limmattal and is planned with a width of about 4.8 metres, which allows safe two-way traffic and makes overtaking easier (See figure 3). The

chosen path follows the regional Richtplan, which identifies this corridor as a priority axis for high-quality cycling infrastructure, ensuring strategic consistency with broader regional mobility objectives (Zürich, 2017). At the three intersections with cantonal Hauptverkehrsachsen, grade-separated underpasses are planned. These reduce the number of necessary stops, allow higher travel speeds, enhance safety by further separating cyclists from motorised traffic, and avoid disruptions to road traffic flow. Underpasses were preferred over bridges because cyclists can use the downhill slope when entering the tunnel to generate momentum for the uphill exit, resulting in lower physical effort and minimal loss of speed. As a possible further enhancement five additional underpasses at smaller crossings could be considered. These would remove remaining conflict points and eliminate stop requirements, thereby further improving the cycling experience if their added benefits outweigh the associated costs. Altogether, the corridor is expected to increase average cycling speeds , improve perceived and actual safety, and make cycling a more attractive option for medium-distance commutes in the Limmattal. The way these elements are phased and implemented over time will be discussed in the next section, as this affects both the costs and the long-term performance of the corridor.

Figure 3: Routing of the bike highway following the proposed route in the regional Richtplan Limmattal with locations of underpasses. Made in QGIS



2 Methods

In this second and analytical chapter, a dynamics Cost-Benifit Analysis (CBA) is made for an uncertain and adaptive future. In this report, a classical CBA is expanded upon by the inclusion of both a Monte Carlo simulation and a modal split determination. After deciding a standard objective function to compute discounted costs and benefits, input variables are sourced. After which, some key variables are randomized using the Monte Carlo simulation, in this way simulating stochastic processes related to unknown future developments. Consequently a modal split is determined using discrete modeling with utility using a logit-function. This computes probabilities for mode use. This mode use and different scenarios of the Monte Carlo simulation are in itself inputs to the otherwise classical CBA. This methodology is described in this chapter, yet inputs data from both the appendices and the model itself, created and submitted in Excel.

Firstly, in 2.1, the different adaptive stages of the system are defined, while in 2.2 scenarios for the management of the infrastructure based on those stages are discussed. In 2.3, input variables are stated, and the choice for a Monte Carlo simulation is motivated, highlighting the stochastic and risk-incorporating nature of infrastructure planning. The model is explained in 2.4 and 2.5, beginning with the base for the modal split and leading up to quantifiable costs respectively.

2.1 Stages

To assure the infrastructure meets needs of the bicycle highway, infrastructure can be (planned to be) adapted. In this way, a plan is constructed to incorporate the uncertainty of future developments and adjust accordingly. Three stages are distinguished:

- Stage 0: The baseline scenario of no intervention. No infrastructure intervention is made.
- Stage 1: Roll-out of the bike highway from the city border of Zurich to the cantonal border in Schlieren. Resulting in a length of 8.5 kilometers and a width of 4.8 meters. Concurrently, underpasses are built at the three major street crossings.
- Stage 2: The Bicycle highway is expanded with five additional underpasses at minor intersections.

Separating the additional infrastructure measures into two stages reflects the hierarchy

between the minor and major crossings and seeks to prioritize spending.

To calculate the cost of the stages, firstly, the cost of the bicycle highway in its total length has to be calculated. As the proposed highway in this report closely follows the routing studied in the cantonal CBA and the width is the same, the cost estimates can be partly adopted. These costs amount to CHF 12.7 million for the stage in between the city border and the golf park in Unterengstringen, as well as CHF 2.1 million for the two additional bridges over the Limmat river (Kanton Zürich, 2017). As the routing deviates after the golf park from the cantonal CBA, the costs must be assumed based on the cost per meter for the given estimate up to the golf park, which measures 5 kilometers. Therefore:

$$\text{CHF 12.7 million/ 5000 meters} = \text{2540 CHF/m}$$

The route from the golf club down to the cantonal border measures 3 kilometers in QGIS:

$$\text{3000 m * 2540 CHF/m} = \text{CHF 7'620'000}$$

This yields results totaling CHF 22.42 million. However, because these cost estimations were published in 2016, they need to be adjusted for the price developments that have occurred in the meantime. The federal building price index lists a price increase of 119.6% in the time frame 2016-2025 for the canton of Zurich (BFS, 2025). Therefore:

$$\text{CHF 22.42 million * 119.6 \%} = \text{CHF 26.81 million}$$

Now the cantonal CBA calculates an additional 10% on building costs to account for purchasing land in the Limmattal. This would amount to:

$$\text{CHF 26.81 million * 10 \%} = \text{CHF 2.681 million}$$

Moving over to the underpasses, it is assumed that they all cost the same, regardless of their status as minor or major crossings and have a length of 20 meters without ramps. Astra published a cost catalogue for foot and bike traffic infrastructure. The cost per meter underpass is CHF 15'000, one ramp costs CHF 90'000, both with a Zurich-specific correction factor of 0.95 (ASTRA, 2010). As these estimates are from 2010 they again must be adjusted to today prices by: 117.9% (BFS, 2025). Therefore per underpass:

$$\text{(20* CHF 15'000 + 2 * CHF 90'000) * 0.95 * 1.179} = \text{CHF 537'624}$$

Thus the total construction cost for Phase 1 in year 1 is:

$$\text{CHF 26.81 million} + 3 * \text{CHF 537'624} = \text{CHF 28'422'872}$$

And for phase 2 in year 1:

$$5 * \text{CHF 537'624} = \text{CHF 2'688'120}$$

Land price in year 1:

$$\text{CHF 2.681 million}$$

2.2 Scenarios

Resulting from the previously stated stages, different scenarios for their implementation can be constructed. These are subsequently tested with a quantitative analysis (later on in the report), for their efficiency regarding the uncertainty of future developments. The analysis is conducted over the course of 40 years because, even though this duration exceeds the lifespan of some infrastructure components, nonetheless the bike highway is expected to have long lasting spatial relevance. Five scenarios are tested:

- **Baseline:** The system stays in an unchanged state, stage 0, for the entire 40 year span
- **Traditional:** Merely the initial infrastructure intervention, stage 1, is made.
- **Robust:** The expanded infrastructure, stage 2, is completed immediately (year 1), to buffer the most extreme future developments.
- **Staged:** Initially, an infrastructure intervention, stage 1, is made. After a certain period, 20 years in, regardless of developments, the additional adjustment of stage 2 are implemented.
- **Flexible:** While initially an intervention, stage 1, is made, the person kilometers for biking is closely monitored. After the this parameter reaches an increase of 77.7% over the initial , stage 2 is implemented.

The switch parameter for the flexible scenario was oriented on four national scenarios outlined by (Schweizer Eidgenossenschaft, 2021). As they have different predictions regarding the increase of bike passenger kilometers until 2050, the switch parameter seeks to estimate whether the increase in bike Pkm in the model follows at least the second highest national scenario. As this scenario has an increase of 77.7% over 33 years.

Therefore, stage two will trigger after an increase of 77.7% in bicycle Pkm, after which there is a lag of three years to model political processes and building permit approval.

The model described in 2.4 and 2.5, quantifies the impact of different choices regarding the bike infrastructure with the use of the societal costs related to the infrastructure. However, the complexity in such a broad system is large, and thus encapsulates a large amount of risk and uncertainty. To capture this risk into the model, a Monte Carlo simulation is ran. This includes predicted key parameters (discussed in 2.3), which development expected to develop, yet are unsure in which (exact) manner. Given this scope of this exercises, this is limited to a total of 3 parameters, the choice for this is motivated in 2.3 as well.

The Monte Carlo simulation draws a random percentile value for each different parameter. This percentile correlates with a given value based on the predicted development of the chosen parameter. Repeated over a large quantity, the Monte Carlo simulation creates a spread for the outcome of an intervention scenario, including extreme uncertain cases. The use of this method highlights the robustness of different scenarios for extreme events for one, or a combination of many, parameters. The Monte Carlo simulation is performed 100 times, on which scenario alterations are performed, a total of 500 outcomes.

2.3 Objective function

In order to quantify results, an objective function is determined. This objective function aggregates all costs related to the infrastructure on a yearly basis. The costs (C_y) are denoted positive and the benefits (B_y) as a negative. The total balance I_y thus constitutes:

$$I_y = C_y - B_y$$

Yearly costs are discounted to a net-present value (NPV) using a discount rate r of 2 % per year. The outcome of all scenarios are thus aggregated using the following formula (starting at $y = 1$):

$$NPC = \sum I_y * (1 + r)^y$$

Except for the baseline scenario, the NPC-values are compared nominally to the baseline result. The baseline is thus a value of 0. In this way, for example, saved travel time costs are not the cost of the total value of time traveled in the region, but the actual saved costs of the intervention made.

To extend comprehensibility and legibility of the model, costs are divided into categories and subcategories, as shown in table 3:

Table 1: Cost Categories and Subcategories

Category	**Subcategory**
Travel costs	Car Travel Times
	Biking Travel Times
	Public Transit Travel Times
	Car Travel Costs
Environmental costs	CO_2 Emissions
	Other Emissions
	Accident Costs
	Health Benefits of Cycling
	Noise Reduction
Intervention Costs	Operational Costs
	Land Acquisition Costs
	Construction Costs

2.4 Variables

Based on the categories, characteristic values are compiled to quantify cost-(sub)categories. These are used as inputs of the model and thus largely determined its' outcome. A total of three variables have been subjected to a Monte Carlo simulation: *Travel Demand, Carbon Emissions & Bike utility.*

Travel demand: The ARE (2021) found an increase of 1.4% in overall traffic per year in Switzerland during the time frame from 2011 to 2021. To reflect the high uncertainty in population fluctuation in the Limmattal and individual transport demand, a variance of 40% over the 40 years was estimated. To put this number into perspective, the Schweizer Eidgenossenschaft (2021) highlights a trend change in yearly overall traffic increases during the two time frames 2000-2009 and 2011-2021 of 1.7% and 1.4%, respectively. Just considering the variance of this recent historic trend change would roughly amount to 2*(1 - (*ratio*))**time* = 70%. In which: *ratio* = 1.7%/1.4%, multiplied by 2 two account for possible positive and negative change and *time* = 2, because the time frame of 40 years is twice as long as the 20 years the change took place in. This means that the assumed

value of 4% is realistic.

Carbon Emission: As the mean value for fuel consumption was found to be 0.139 kg/Pkm (Kanton Zürich, 2017). The yearly reduction was calculated from around 0.18kg/Pkm in 2000 to 0.12kg/Pkm in 2015 (Fontaras et al., 2017). This equals a yearly fuel consumption reduction of 0.004kg/Pkm. The variance of this value seems to be high especially as the focus of research shifts to electric vehicles and they may be less development on such motors. Vice-versa the need for efficient petrol motors could strongly increase through climate change, leading to increased development. Therefore, a variance of 40% is assumed.

Bike utility: The mean bike utility value of -1.87 and its yearly reduction are taken from the excel model provided. The variance value was oriented towards four different scenarios for the development of traffic in Switzerland for 2050 (Schweizer Eidgenossenschaft, 2021). The scenarios forecast an increase in the bicycle model split between 29.8% and 110.4% from 2017 to 2050. This equals a spread of 40.3% from the center of the two extreme scenarios. Still, bike utility (the societal preference towards bikes) and model split development are not the same; their uncertainties are largely related; therefore, 40% is chosen.

Table 2: Transport Mode Data Parameters. Values in bold are uncertain.

Car		
Variable	**Value**	**Description**
Fuel Consumption	0.06 L/km	Average fuel consumed per kilometer driven in China (UVEK, 2024)
Fuel Cost	1.59 CHF/L	The average fuel price per liter for cars (Schweizer Eidgenossenschaft, 2024).
Carbon Emission production	**0.139** $KgCO_2/km$	Kilograms CO_2 emissions for cars per kilometer, uncertain (Kanton Zürich, 2017).
Carbon Emission cost	0.04168 CHF/Kg	Social cost per kilogram of CO2 emitted (Kanton Zürich, 2017).
Traveltime Cost Car	23.3 CHF/h	The monetary value of one hour of travel time for car users (Kanton Zürich, 2017).
Travelspeed Car	32.4 km/h	Average travel speed for cars (Bösch et al., 2018).
Accident Rate Car	0.078 CHF/pkm	The average accident cost per person-kilometer (Kanton Zürich, 2017).

Table 2: Transport Mode Data Parameters. Values in bold are uncertain.

PM10 Emission	0.04 g/pkm	Emission factor for PM_{10} particles (Kanton Zürich, 2017).
NOX Emission	0.132 g/Pkm	Emission factor for nitrogen oxides (Kanton Zürich, 2017).
Zinc Emission	0.0014 g/Pkm	Emission factor for zinc (Kanton Zürich, 2017).
PM10 Emission Cost	0.276 CHF/g	Societal cost of PM_{10} per gram (Kanton Zürich, 2017).
NOX Emission cost	0.0033 CHF/g	Societal cost of nitrogen oxides per gram (Kanton Zürich, 2017).
Zinc Emission	0.0114 CHF/g	Societal cost of noise pollution (Kanton Zürich, 2017).
Noise pollution passenger car	0.0116 CHF/pkm	Societal cost of noise pollution (Kanton Zürich, 2017)
Petrol reduction in taxes	0.87 CHF/L	Ratio of petrol consumption (Kanton Zürich, 2017).
Bicycle		
Traveltime Cost Bicycle mixed traffic	24.80 CHF/h	The value of one hour of travel time for cyclists in mixed traffic (Kanton Zürich, 2017).
Traveltime Cost Bicycle seperate lane	18.70 CHF/h	The value of one hour of travel time for cyclists in mixed traffic (Kanton Zürich, 2017).
Health Benefit Bicycle	1.306 CHF/Pkm	A medium monetized health benefit per kilometer cycled in the literature (a negative cost).(Kanton Zürich, 2017).
Average Travel speed mixed traffic	13.00 km/h	The average speed for cycling if in normal traffic (Kanton Zürich, 2017).
Accident Cost mixed traffic	1.02 CHF/Pkm	The average cost of accidents per person-km for cyclists (Kanton Zürich, 2017).
Public Transport		
Traveltime	See 7	The travel time spent in the vehicle between origin (i) and destination (j).
Fare Cost	See 6	Cost of using public transport for each origin and destination connection. As there is much political debate about these prices, this variable is uncertain.

Table 2: Transport Mode Data Parameters. Values in bold are uncertain.

Value of Travel Time (PT)	14.40 CHF/h	The monetary value of one hour of travel time for PT users. (Kanton Zürich, 2017).
Access time	**6.00 min**	The average time spent getting to and from a PT stop for a trip. As the usual capture area for buses is around 500 meters, the time is taken to walk this far. As this value can change in the future through a denser network it is considered uncertain.
Accident Rate (PT)	0.024 CHF/Pkm	Average accident cost per Pkm.

Table 3: Intervention Data & Parameters

Stage 1		
Variable	**Value**	**Description**
Length	8.50 km	Total length of the bike highway as measured in QGIS
Width	4.8 m	Width of the bike highway, as adopted from the cantonal cost benefit analysis (Kanton Zürich, 2017).
Utility improvement	1.50%	A coefficient for the increased utility of cycling due to the new infrastructure.
Distance Reduction	5%	The percentage reduction in travel distance for cyclists using the new route.
Bike Speed on bike highway	16.00 km/h	The cantonal CBA lists average travel speed on a bike highway as 20 km/h, but as there are still many obstacles on route, this value is decreased to 16 km/h (Kanton Zürich, 2017).
Accident Cost separate lane	0.10 CHF/Pkm	The average cost of accidents per person-km on the bicycle highway and built underpasses (Kanton Zürich, 2017).
Construction cost	28'422'872 CHF	Construction cost of the bike highway consisting of the length and width as well as major underpasses (See 2.1).

Land cost	2'681'000 CHF	Cost of land acquisition (See 2.1).
Stage 2		
Utility improvement 1	2.50%	A coefficient for the increased utility of cycling due to the new infrastructure.
Bike Speed on bike highway	18.00 km/h	The average speed increases by 2 km/h after completion of the additional underpasses.
Accident Cost (Stage 2)	0.05 CHF/Pkm	The average cost of accidents per person-km after completion of the additional underpasses. The former value was halved as the remaining dangerous sections.
Construction cost	2'688'120 CHF	Construction cost of the five minor underpasses (See 2.1).
Other cost		
Operational cost	3.84 CHF/m^2	Operational cost of the bike highway (Kanton Zürich, 2017).

2.5 Utility function

In order to estimate possible future modal splits for different scenario, modifiable utility functions are used to characterize benefits (or disadvantage) of travel. Actors in the system are assumed to be generally rational and aim to optimize this utility (U_k) in their mode choice. Utility functions have the following form:

$$U_k = \gamma_k + \sum \beta_{ik} * x_{ik}$$

In this formula, the utility U_k is determined by a mode-specific constant γ_k for utility (inherent preference) and the sum of the utility derived by all attributes $\sum \beta_{ik} * x_{ik}$. Fuel cost, waiting times and general travel times are examples of these attributes, where the utility scaling coefficient β_{ik} for, for example, travel time might differ per mode. Utility constants are given in table 3:

Table 4: Utility Function Parameters

Name	**Value**	**Description**
Travel Time Coefficient car	-7.87	The weighting factor for car travel time in the utility function.
Travel Time Coefficient bike	-6.80	The weighting factor for bicycle travel time in the utility function.
Travel Time Coefficient PT	-2.74	The weighting factor for public transport travel time in the utility function.
Utility Constant Car	-1.53	A constant representing the inherent preference or bias for car travel.
Utility Constant Bike year 1	**-1.87**	A constant representing the inherent preference or bias for cycling.
Utility Constant PT	-0.79	A constant representing the inherent preference or bias for public transport.
Fuel Coefficient	-0.34	Weighting factor for fuel cost in the car utility function.
PT Fare Coefficient	-0.34	Weighting factor for public transport fare in the utility function.
PT Access Time Coefficient	-5.32	Weighting factor for time spent accessing public transport.

Consequently, utility functions are translated into probabilities of choosing a certain mode using a logit functions, given by the following formula:

$$P_k = \frac{e^{U_k}}{\sum e^{U_k}}$$

These probabilities are in itself the modal split, where the probability is better viewed as the probability a random person would choose a certain mode, instead of a person choosing a mode randomly. The logit-function introduces 'realness' into the model, representing differing taste in the population, uninformed choices or randomly different choices by agent (people) in the system.

3 Results & Discussion

The model set up in chapter 2 was run a hundred times for the 5 possible scenarios. The result is a broad overview of the estimated 1st to 99th percentile value outcomes with respect to NPV, as can be noted in table 5.

3.1 Baseline Scenario

The baseline scenario represents the development of the transport system without constructing the bike highway. Across all iterations, the model shows a steady increase in total societal costs over the 40-year horizon (see Table 5, primarily driven by rising travel-time-related costs as overall travel demand grows in most possible futures while infrastructure capacity remains unchanged. A slight increase in cycling can be observed due to the exogenously improving bicycle utility, yet cyclists remain in mixed traffic at low speeds and high accident rates, which results in growing accident-related costs that outweigh the associated health benefits. Environmental and externality-related costs also rise, reflecting the combined effect of increasing person-kilometres travelled (pkm) and the continued dominance of motorised modes. Overall, the baseline results illustrate that maintaining current infrastructure conditions leads to progressively higher societal costs, and therefore provides a clear reference against which the benefits of the intervention scenarios can be evaluated.

3.2 Traditonal Scenario

In the Traditional scenario, only the first stage is realised, providing a continuous bike highway between Dietikon and Altstetten with three underpasses at intersections with *Hauptverkehrsachsen*. This partial intervention already produces a noticeable improvement compared with the baseline (see figure 4). Travel-time-related benefits form a large contribution: cyclists travel at higher and more consistent speeds on the dedicated section, and a moderate share of car users shifts to cycling due to the improved perceived utility of the mode. Because the value of time and fuel costs for car travel are comparatively high, this modal shift translates directly into substantial monetary gains. Importantly, the resulting travel-time savings alone exceed the investment costs of Stage 1 for some iterations, meaning that even without considering safety or environmental effects, the intervention is already generally socially beneficial. Safety- and health-related components further reinforce this

outcome and are largely the biggest driver of the NPV outcome. Accident costs fall as more trips occur on separated infrastructure with significantly lower per-kilometre risks than mixed traffic, while increased cycling generates additional health benefits. Classical environmental improvements such as reduced emissions and noise also occur as car travel is substituted by cycling, though their contribution remains small compared with the dominant accident and health components in the model. Overall, the Traditional scenario demonstrates that even a partial realisation of the bicycle highway delivers clear societal benefits, driven primarily by travel-time savings and safety improvements as a result of separating modes.

3.3 Robust Scenario

In the Robust scenario, both stages of the bicycle highway are implemented immediately in year one, resulting in a fully separated corridor with additional underpasses at the remaining relevant intersections. This comprehensive intervention further amplifies the improvements observed in the Traditional scenario (see figure 7). Travel-time-related benefits remain the dominant component and increase substantially: cyclists travel at higher and more stable speeds across the entire route, and a stronger shift from car to bicycle occurs due to the further improvement in cycling utility. Because car users are highly sensitive to travel-time differences and face comparatively high monetary values of time and fuel costs, this additional modal shift generates considerable gains. On average, the resulting travel-time savings are roughly twice as large as in the Traditional scenario and already exceed the combined investment costs of both stages, meaning that the expanded infrastructure is socially beneficial even when considered purely from a travel-time perspective. Safety- and health-related effects reinforce this outcome. The additional underpasses reduce the number of conflict points along the corridor, lowering per-kilometre accident risks even further, while the higher volume of cycling enhances health benefits. The incremental improvement in these components, however, is smaller than the initial jump from the baseline to the Traditional scenario (compare figure 4 and 5). This is because the main decline in accident costs stems from the shift from mixed-traffic cycling to dedicated infrastructure in Stage 1, whereas the additional reduction from removing remaining intersection conflicts in Stage 2 is comparatively limited in scale. Emission and noise pollution effects also improve as more trips shift away from the car, but their contribution remains secondary relative to safety and health effects. Overall, the Robust scenario shows that implementing both stages from the start produces pronounced societal benefits. It must be noted that the improvements are realized regardless of traffic volume in this model, such that the additional increases in speed or health benefits are benefited from instantaneously.

3.4 Staged Scenario

In the Staged scenario, the bicycle highway is first implemented with the Stage 1 infrastructure, while the additional underpasses of Stage 2 are introduced only after twenty years. Because the full highway becomes available later, the associated benefits are delayed compared to the Robust scenario. Travel-time savings and modal shift remain similar to the Traditional scenario during the initial period and increase only once Stage 2 is added, meaning that a substantial share of these gains arises relatively late and therefore carries less weight in discounted terms. The same timing effect applies to safety and health benefits. From an investment perspective, the staged approach spreads costs over time, and the investments for Stage 2 in year 20 are discounted, reducing its weight in the net-present-cost calculation. Nevertheless, the analysis indicates that earlier implementation of Stage 2 (Robust Scenario) would still generate greater benefits because the high returns from qualitative infrastructure improvements begin to accrue sooner. Overall, the Staged scenario demonstrates that postponing the second phase diminishes the scale and timing of benefits, as it can be noted to be a broad average of the traditional and robust model in figure 7. While it still delivers positive societal outcomes, the Robust scenario remains more favourable, as its earlier investment enables the substantial benefits of the full corridor to accrue sooner and outweigh the higher upfront costs.

3.5 Flexible Scenario

In the Flexible scenario, the second stage of the bicycle highway is implemented only once the number of kilometers cycled increases by 77.7%, with an additional three-year delay after this threshold is reached. Across the simulations, this leads to a substantially later introduction of Stage 2 compared with the Staged scenario, and in some iterations it occurs very late within the 40-year horizon. The average switch occurs around the 25-years mark, and in around 35 % of cases no switch happens at all. As a result, travel-time-related benefits remain close to the Traditional scenario for a prolonged period and only begin to increase once Stage 2 is eventually activated. Consequently, the overall travel-time savings fall below those of the Staged and Robust scenarios, reflecting both the late switch and its reduced discounted impact. Modal shift follows the same pattern: cycling volumes grow more slowly and only rise further once the infrastructure upgrade is finally implemented. Because the improvement arrives so late, the Flexible scenario captures almost none of the additional safety and health benefits associated with Stage 2. Although the delayed investment is more strongly discounted (see figure 6), this cost advantage does not offset the loss of early benefits. The high threshold for triggering Stage 2 thus leads to an

implementation that is later than socially optimal. Nevertheless, the flexible approach may hold practical relevance where political or public support depends on demonstrating sufficient demand before committing to major infrastructure upgrades. Overall, the Flexible scenario performs worse than approaches where Stage 2 is implemented earlier, illustrating the societal cost of delaying improvements that offer high returns once in place.

3.6 General Remarks

Overall, the robust scenario was the one with the most positive cost benefit analysis, while all scenarios were positive in relation to the baseline one and have a positive cost benefit ratio. Regarding the implementation of phase 2, this means that it's modeled benefits are so positive that they outweigh the discounted infrastructure costs associated with later implementation. Therefore, the model suggests that all infrastructure should be built at once. However, owing to the large uncertainties in travel demand and travel behavior (with respects to bike utility), the difference between iterations vary immensely. This points to the dependency the success of the project has on the influx of new travelers by bike, either because of growing demand or shifting mode preference. Furthermore, looking at the standard deviations within the iterations of the model, they all, except the baseline, hover around +-35% of the average. This variance is a propagation of the uncertainties of 40% introduced in the Monte Carlo simulations. The average modal split for biking in year 40 was over 20% in the baseline, 30% in the traditional, and 35% in the robust scenarios. The total saved cost range of CHF 347 million in the traditional up to CHF 482 million in the robust scenario compared to baseline seems probable. The Kanton Zürich (2017) has a range of around CHF 180 million and CHF 900 million. The calculated values sit inside this range. The mean values per category are noted in table 5, and the spread of NPV per iteration and scenario in Figures 4 to 7.

Table 5: Modelled mean results of the five scenarios. Absolute values for the *Baseline* model, others have this as reference value. Averages over the 100 iterations. [CHF million]. TTT: Transport travel time cost, ENV: Environmental cost, INV: Infrastructure cost, Total: Total cost.

Scenario	**TTT**	**ENV**	**INV**	**Total**
Baseline	1323.3	32.6	0	1355.9
Traditional	-30.696	-352.765	35.475	-347.986
Robust	-61.047	-459.454	38.164	-482.337
Staged	-46.904	-405.287	37.284	-414.907
Flexible	-41.665	-391.163	36.494	-396.334

Figure 4: Share of iterations which reached a specific transport travel time cost [CHF million]. Excel output.

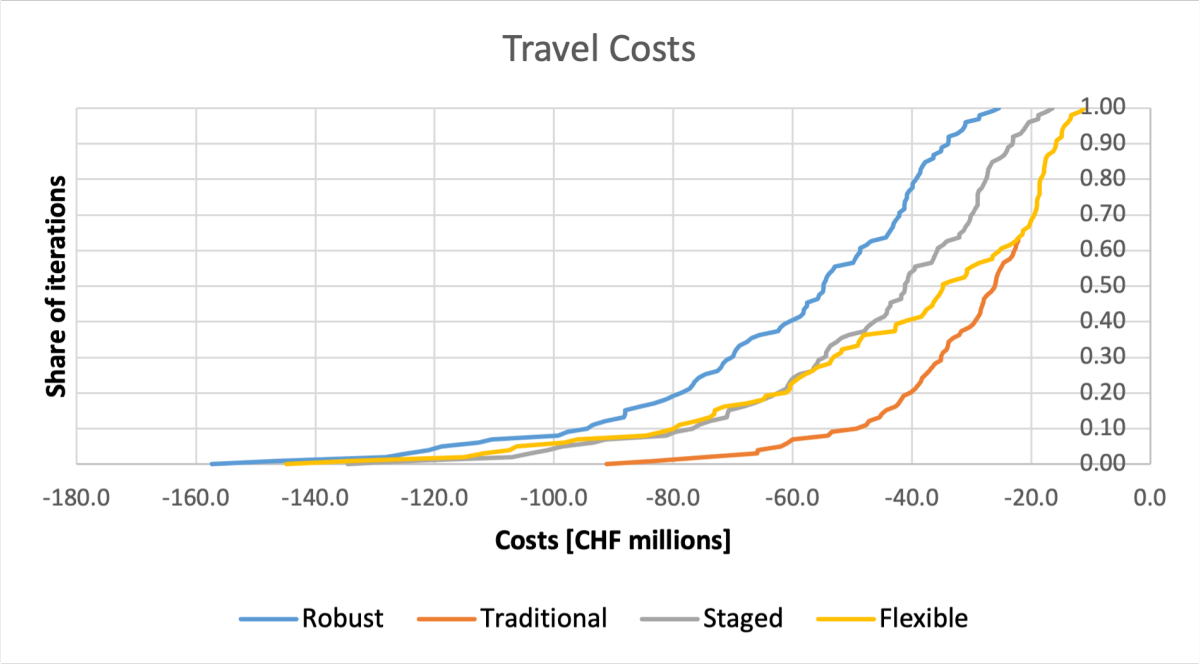


Figure 5: Share of iterations which reached a specific environemental cost [CHF million]. Excel output.

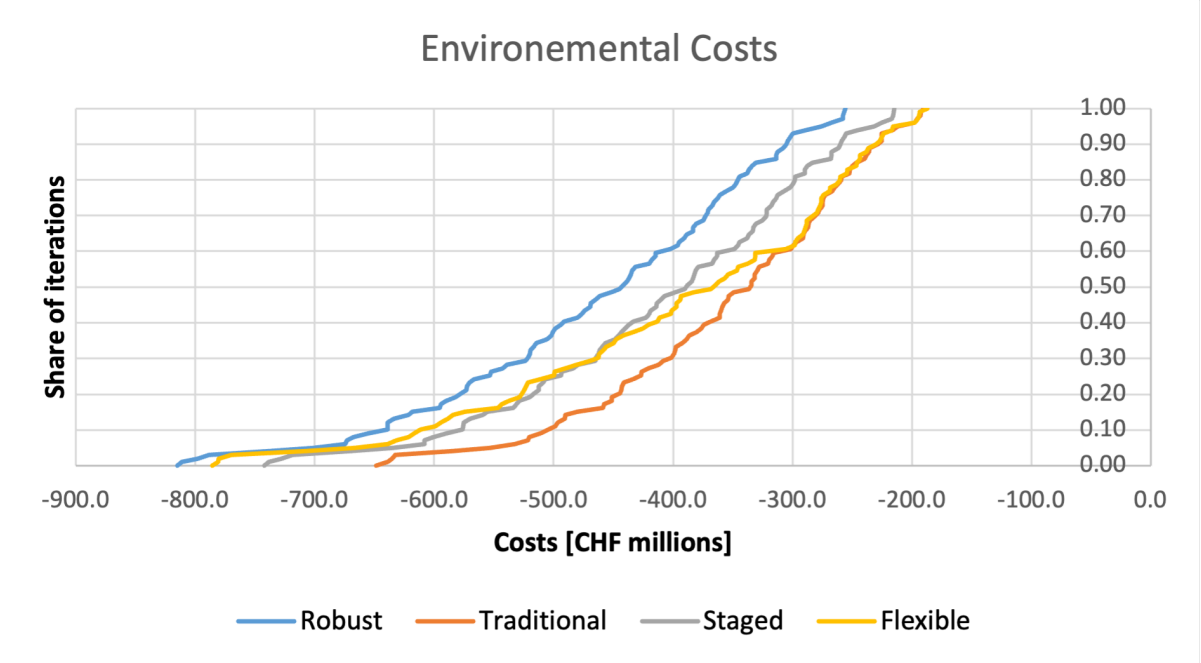


Figure 6: Share of iterations which reached a specific infrastructure investment cost [CHF million]. Excel output.

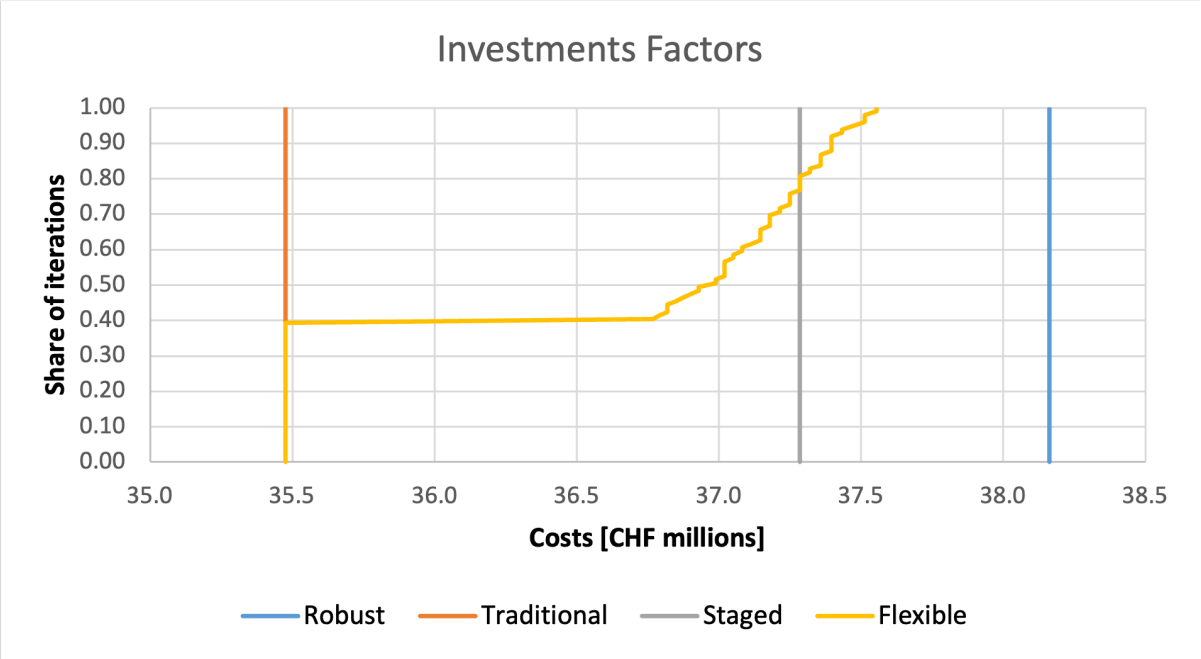
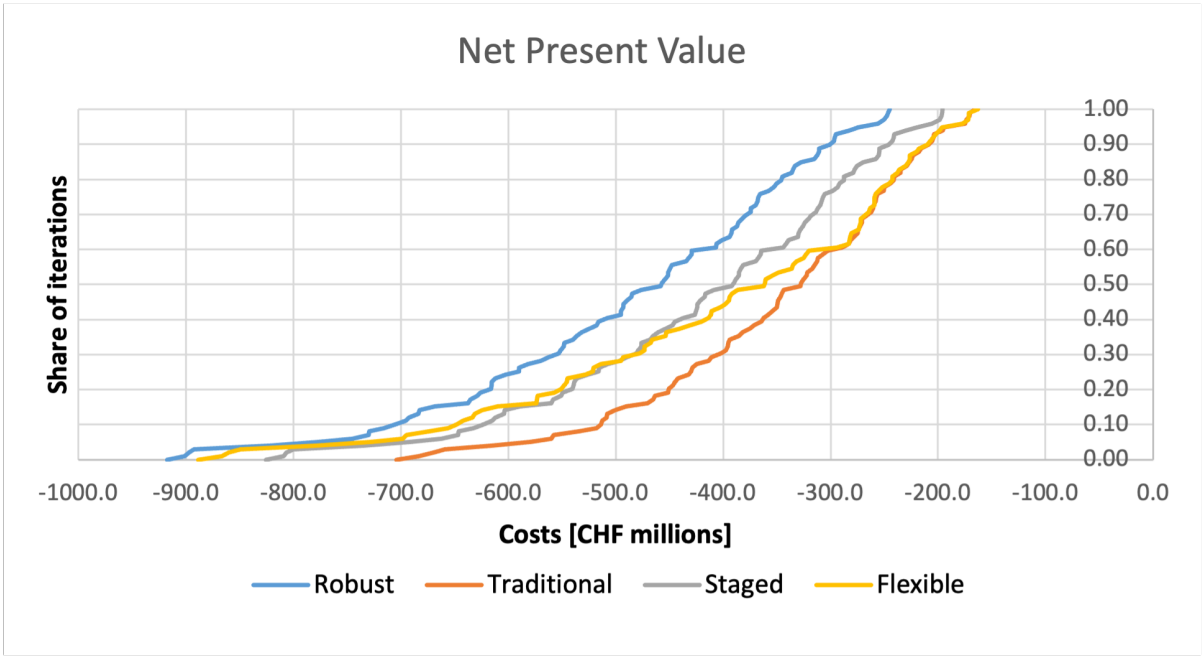


Figure 7: Share of iterations which reached a specific total cost [CHF million]. Excel output.



4 Conclusion

Across all scenarios, the analysis demonstrates that constructing a bicycle highway in the Limmattal generates substantial societal benefits compared with maintaining the status quo. Even the more limited interventions yield clear net gains, driven primarily by improved travel-time performance and enhanced safety conditions for cyclists. Among the evaluated strategies, the Robust scenario, where both stages are implemented at the outset, emerges as the most favourable, as its earlier delivery of high-quality infrastructure enables benefits to accrue sooner and at a greater magnitude. These findings underscore the relevance of timely investment in cycling infrastructure as an effective means to support efficient, safe, and sustainable mobility in the region. While the results consistently find early and full implementation to be preferable, critically examining the outcomes and comparing them to the Kanton Zürich (2017), highlights a number of limitations. Firstly, the mean model splits are too high to be feasible. the Kanton Zürich (2017) found that a bike modal share of over 20% in the Limmattal is not likely, even with the bicycle highway. Our model meets that threshold even in the baseline scenario in year 40. This could mean that the selected bike utility increase may be too optimistic. Regarding the composition of the total costs, the environmental costs are approximately ten times larger than the travel time and infrastructure costs in the non-baseline scenarios. This is caused by the very high valuation of health benefits and accident cost reduction. The health benefits value of 1.306 CHF/Pkm proposed by Kanton Zürich (2017) has a strong impact on the model; meanwhile, there is much uncertainty about this value in the literature, which could be considered in an improved model. The accident rate of cycling after the implementation of stage 2 is also very low, sitting at 0.05, which is between car at 0.08 and public transport at 0.02 CHF/pkm. The chosen value seems too low and has to much benefit. Meanwhile, the other components of environmental cost constitute a minor share, which means that greenhouse gas and smog emissions are not valued highly in the model. Reducing the skewed impact of health benefits and accident costs, the valuation of the emissions could be increased. This would represent a more environmentally conscious model. Lastly, an important limitation of the model was the lack of any restrictions on increased biking. This means that bicycling is always positive and there is no congestion, as the robust scenario has the highest cost/benefit. Introducing a capacity limit for the bike highway would therefore balance the model.

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A AI Statement

The use of generative AI, specifically GPT-5.1, was used for minor task in the creation of this report. It was used in the following manners:

1. Rephrase and shorten texts where needed
2. Tidy tables in the Overleaf interface
3. Debug function in the Excel file

These uses were merely supportive and no creative or executive authority was given to AI in any form.

B Stakeholder Analysis

In addition to fitting into a larger framework and being financially viable, the Veloschnellbahn also aims to fulfill the needs and demands of certain stakeholders. These needs are the basis for the larger national and cantonal frameworks, as popular opinion for example was the leading cause behind the "Bundesbeschluss über die Velo-Initiative", such that this project should fulfill those higher-level needs. Besides, such needs are critical for further development of the project planning on a smaller scale in SIA Phases 2 & 3. As of the writing of this report, no such publicly available stakeholder analysis exists, such that the following is based on publicly available goals and involved parties. Internally, there is evidence a stakeholder analysis has been done by consultant TBF-Partner (Veloschnellroute Limmattal — TBF + Partner AG, z.d.).

Firstly, all stakeholders are compiled and displayed in a Power-Interest (PI) matrix (see figure 9), as per a classical stakeholder analysis. Using this matrix, an internal proposal on communication strategy and stakeholder management can be made, see Figure 10. Subsequently, effective groups can be distinguished.: Approving, Shaping, Participating and Controlling. Such a distinction can highlight power dynamics within the project.

Afterwards some stakeholders, precisely the ones most directly connected to needs for further project detailing, are divided in four categories: Users, Owners, Directly Affected Public and Indirectly Affected Public, where this distinction is made to differentiate

how these needs are differently incorporated. Stakeholders' are discussed with regards to Needs, actionable things with their desire and benefit with respect to the completion of the project, and Concerns, negative effects of the projects which for their approval need to be mitigated or discussed (on why not to engage in mitigation). Needs and Concerns can be used to define both actionable goals (e.g. CO2 reduction) as design parameters (Bike Path width).

B.1 PI Matrix & Communication Strategy

In a complex project, encompassing multiple levels and scales and reaching most parts of the general public, a large number of stakeholders can be imagined. Yet, for the scope of this report, it has been limited to a handful which are either most important, or deemed to illustrate the complexity and different involved parties in this project. They have been outlined in figure 10, in a Power-Interest matrix (PI). This aims to align the Power a stakeholder has to shape and influence the development and the project, and the Interest it has with it's successful completion.

Power can be evaluated in terms of administrative control, or pushing power over executing authorities. ARE & ASTRA for example, has large influence over the national policy, shaping and piloting the initiatives following the Veloweggesetz. It can contribute up to 35 % as part of the nationally-steered Agglomerationsprogram Limmattal, and thus holds significant power over the affordability and thus the execution of the project.

Interest concerns the need for the execution of the project, and its' interest in the further detailing of the project. High interest parties can be strongly impacted by the manifestation and success in later stages of the SIA-stages. Bikers, as users, are high-interest stakeholders, as if they are not satisfied, and usage of the bike highway is not seen as a preferable option, the project efficiency is flatlined. With further detailing of the project, in order for successful implementation, their needs need to be incorporated, whereas lower interest groups might offer valuable public opinion which is useful on a higher level, their feedback is not (necessarily) vital for project succes.

Based on the PI-matrix, broad terms for communication strategies for internal and external management can be distinguished, which is part of SIA 111. 4 simplified categories are noted in figure 11 and deem as a guide for the project executor, in this case, mainly the Canton of Zürich (or Aargau depending on the stretch) in cooperation with municipalities. For example, the actively collaboration with Nearby residences, commercial establishments and offices is important for fitting in the bike highway on a granular level, while keeping public support on the executors side.

Figure 8: Power-Interest matrix

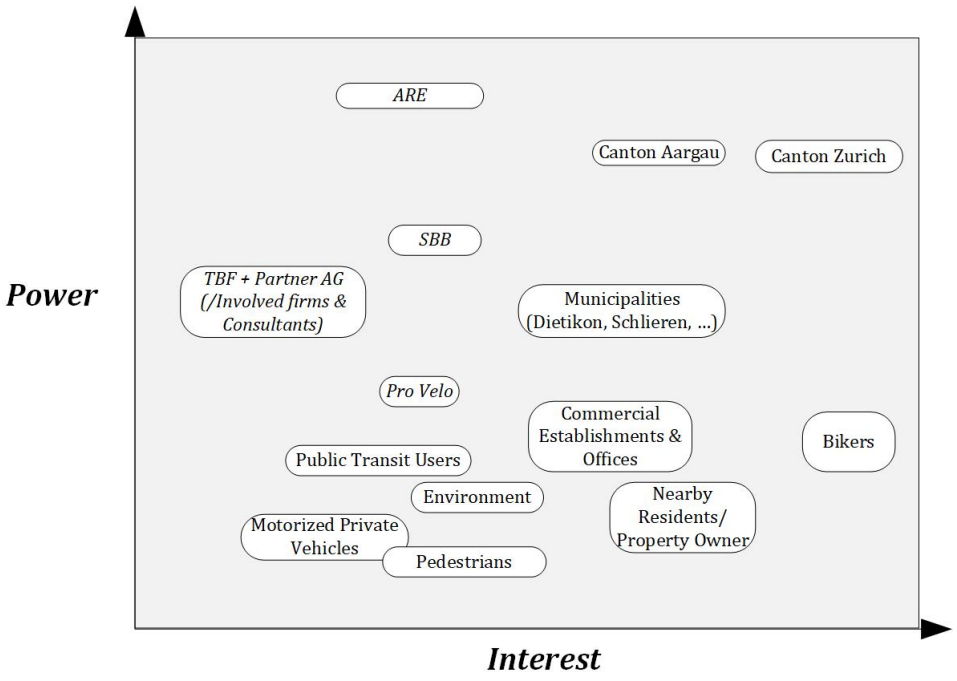
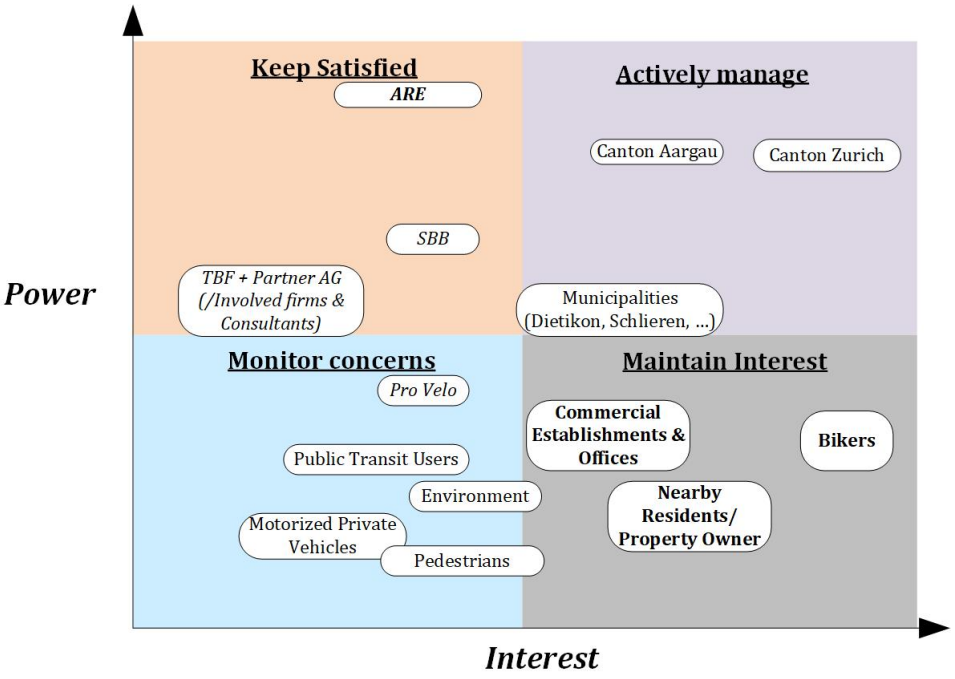


Figure 9: PI-matrix with communication strategies



B.2 Functional Groups

The communications strategies do give little inside into the bureaucratic structure of the project, such that a new distinction, functional groups is made, based on course content by Elvarsson. Approving, shaping, participating and controlling stakeholders serve a different and important function for the stepwise execution of the project. Participating groups, such as SBB or Consultants are important for execution of parts such as land acquisition, site analysis or integration in the surrounding areas. On a more top-down level, this piece-wise execution does need to fit into the larger federal and cantonal framework of approving stakeholders. Bottom-up however, the needs and concerns of the parties whose regular lives are affected by the project weigh equally important, especially so in the more bottom-up planning approach in Switzerland. Its' up to the shaping stakeholders to balance this needs, to serve both local and federal agendas and synthesise into a conforming, optimal solution. Besides federal and cantonal frameworks, splitting shaping and controlling stakeholders into relational categories related to the Veloweg, is used in

this report to specify further needs.

Figure 10: PI matrix with functional groups

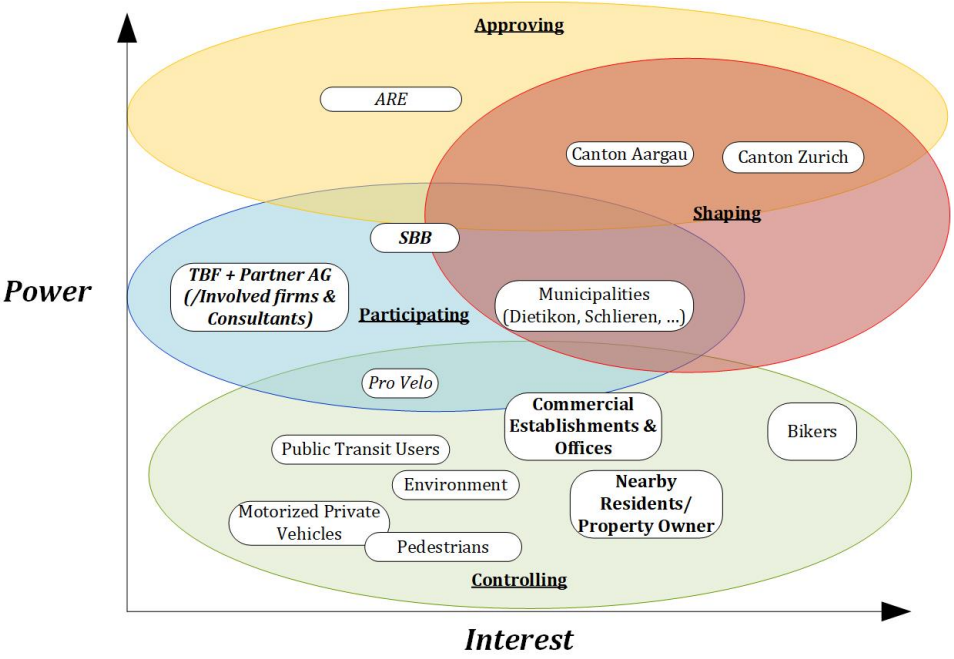
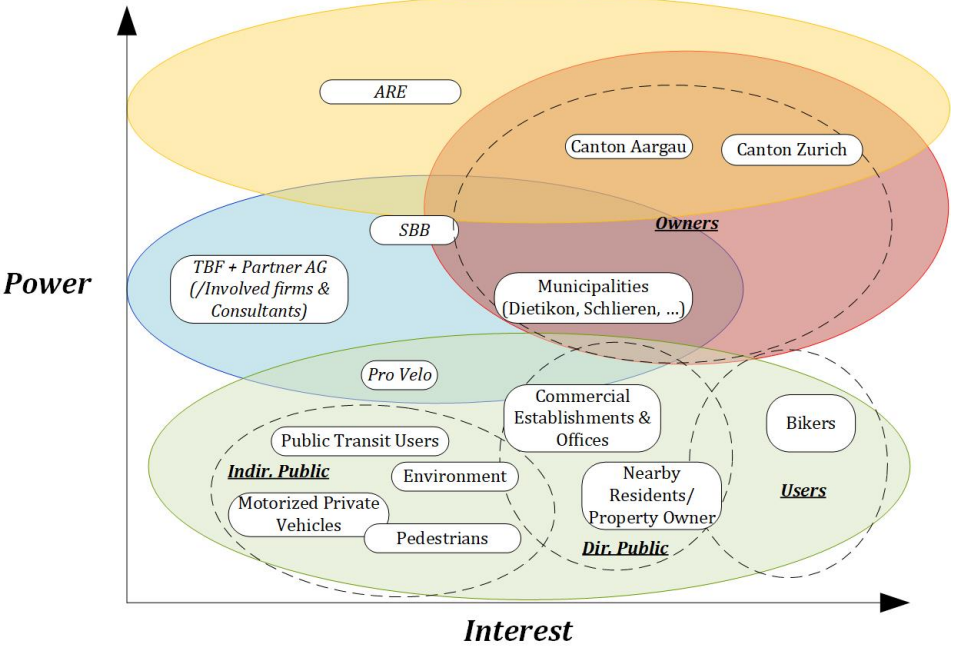


Figure 11: PI matrix split on different needs and concerns



B.3 Categorized Needs and Concerns

Users are defined by their active stake in using the newly-developed infrastructure. The project must meet their needs, as usage is required for positive societal and financial outcomes. The only users of the infrastructure for this project are Cyclists. This includes different modes, such as regular bicyclists and micromobility users (E-bike, electric scooter etc.) and various motives for mobility, such as recreation and commute.

With regards to design this would translate in the following Efficient routing to places people want to travel

Group	***Needs***	***Concerns***
Cyclists	**A fast and safe connection.** This leads to more willing to travel and perhaps travel time saved compared to other mobility types. In addition, safety is important, especially with regard to recreational use	Conflicts with other modes of transport (pedestrians, cars,..) and following safety concerns. Besides capacity constraints might hamper flow, comfort and safety further.

- Fewer Intersections
- Clearer signage
- Ample width and space for overtaking, obstacles and two-way traffic
- Efficient connectors
- Preferred separation from motorized private transport
- Auxiliary infrastructure such as storage space, repair shops

Owners are stakeholders which hold influence and responsibility over the execution and acquisition of land for this project, and such are also responsible for maintenance and adequate operation. The stakeholders which fall into this group are the ones which are part of the administrative system, such as cantons and municipalities. While they both have similar functions, their difference in scale induces different levels of control and needs

Group	***Needs***	***Concerns***

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Canton(Aargau Zürich)	Increase modal split for cycling in accordance to federal objective SOURCE. Improved mobility and modal split for region and a satisfied public opinion	Low financial viability and negative public support decrease incentives for this project
Municipalities (Schlieren, Dietikon,....)	Local growth in accessibility, increasing attractiveness and connections to other areas.	Clashes with local property owners, in addition to general lack of support, can increase barriers to the project.

With regards to the design process this would translate in the following:

- A thorough and positive cost-benefit analysis, which takes into account properly viable alternatives for future adaptations
- Cooperation between levels of government (Federal, Cantonal, Municipal), with regards to larger frameworks and authority in execution.
- Proper communication with groups of the public

Directly Affected Public encompasses parties whose lives are explicitly affected by the execution of the bike highway. This is a strongly spatially bounded group, where the general effect of (construction of) public space is an important factor, where these parties all have different functions within this public space.

Group	***Needs***	***Concerns***
Nearby Residents	Faster travel times and increased opportunities to bike, for both recreation and commute. It can also increase land values and the attractiveness of the neighborhood. Additionally less car traffic in local streets.	The construction can create nuisance. More traffic in the area can disturb communities in the form of noise pollution and high use of public space for transit. Other transport modes might become more inefficient, which could be a problem for non-bike owners. Reduction of parking spaces

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Commercial Establishments	More accessibility for customers by bike. Increased visibility for users in public space. General increase in neighborhood attractiveness benefitting them	Difficulties with accessibility for trucks customers by car, leading to higher operational costs and/or reduced customer base. Reduction of parking spaces
Offices	More accessibility for employees by bike	Difficulties with accessibility for trucks employees by car

With regards to design this would translate in the following:

- Efficiency of the infrastructure should benefit all groups, not only bike users. Prevention of higher travel times by other modes.
- Integration of bike path into public space efficiently, preferably not taking away from pavement space. Parking and loading spaces are most dominant.
- Efficient construction operations and a minimizing of discomfort associated with realisation infrastructure.

The realization of the project will have numerous effects on the complex urban environment around itself, the Indirectly Affected Public. This results in parties or causes being affected in an implicit way, yet it is noteworthy to qualitatively and quantitatively assess these parties for the complete scope of such a project. This could be seen as a summation of large public opinion groups.

Group	***Needs***	***Concerns***
Environment	Less environmental (air) pollution by the reduction of MPT usage. Less energy usage by more bike traffic	Additional construction may induce environmental harm
Motorized private transport	Lower congestion on roads	Reduction in parking spaces and longer travel times for MPT's due to rerouting
Public Transit Users	Higher usage rates since stops are more accessible	The replacement of transit usage by bike traffic

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Pedestrians	Less cars on the road leads to improved safety	Veloschnellbahn-users may endager pedestrians and take away sidewalk space

With regards to design this would translate into proper integration into the urban space and other mobility networks. Additionally, this groups delivers a lot of signs of which quantitative indicators and goals could be established for an appropriate assessment of feasibility and efficiency of a design.

In a complex project as such, project developers need to make decisions which compromise some needs and don't address concerns. There are different possible strategies to deal with unfulfilled needs of stakeholders, beneath explained for the example forthe removal of parking lots:

- **Mitigation**
  Further project phases could, when increasing definition, aim to minimize the parking space loss within the current plan, leaving those where possible, thus minimizing the negative effect.
- **Divertion**
  Concession can be done in the form of promising the placement of alternative parking lots different locations nearby the project, in a more optimal manner.
- **Ignorance**
  The executive parties in the project could use the Velo-Initiative as a ground to stand behind its' intentions to push the modal shift towards bicycles by simply ignoring concerns about reduced parking spaces, making it more difficult for people to use a car, thus more attractive to use a bike (or alternative).

C Important input matrices

Table 6: Public transport fare costs. The value has been calculated through consulting the ticket prices on the SBB website to travel across the zonal boundaries for adults with a Swiss half-fare card.

origin (i)	destination (j)	Fare (CHF)
Altstetten	Dietikon	7
Altstetten	Schlieren	3.5
Dietikon	Schlieren	4

Table 7: Travel time public transport

origin (i)	destination (j)	Traveltime (hr)
Altstetten	Dietikon	0.125
Altstetten	Schlieren	0.092
Dietikon	Schlieren	0.122

Table 8: Demand between origins and destinations

origin (i)	destination (j)	Demand (trips)
Altstetten	Dietikon	6333
Altstetten	Schlieren	7600
Dietikon	Schlieren	5066

Table 9: Distances between origins and destinations (km)

origin (i)	destination (j)	Distance (km)
Altstetten	Dietikon	7.5
Altstetten	Schlieren	3.6
Dietikon	Schlieren	5